

Lesson Title:	2.1 Understanding Language: Interactions with Artificial Intelligence	 Artificial Intelligence For Georgia
Grade:	6th -8th	
Subject:	Computer Science	
Duration:	1 class period	
Objective(s):	Students should know: • Language has evolved as ... culture • NLP • The knowledge that agents are using to respond to speech Students should be able to: • List various ways in which they or people around them use language to interact with computers in their daily lives. • Distinguish concepts such as speech recognition, speech generation, and question answering.	
Depth of Knowledge:	2	
Standards Aligned:	Georgia Standards of Excellence (GSE) for Science: MS-ETS1-3: Analyze data to determine how the variables affect the outcomes of a design problem. Georgia Standards of Excellence (GSE) for Computer Science: CS.CS.6: Develop programs that use conditional statements and/or loops to solve problems. CS.CS.5: Analyze the impacts of computing on society and culture. CSS.CT.6-8.33 Utilize computational thinking to solve problems.	

	CSS.DC.6-8.7 Explore the relationship between computer hardware and software. MS-CS-FSIS-2 Investigate and identify the basic components of computers and networks. MS-CS-FCP-5 Explore the relationship between computer hardware and software.
Materials:	Student Laptops Teacher Slides (has answers covered) Student Work Slides (has blanks for students to fill)
Vocabulary terms and definitions:	Speech Recognition: The ability of a computer or device to identify and process human speech, converting spoken words into text. Speech Generation: The process by which a computer produces spoken language, allowing it to communicate verbally with users. Natural Language Processing (NLP): A field of artificial intelligence that focuses on the interaction between computers and humans through natural language. Smart Assistant: A software application that uses artificial intelligence to perform tasks or provide services through voice commands (e.g., Siri, Alexa). Text Generation: The ability of AI systems to create written content based on input or data, often used in chatbots and automated writing tools. Machine Translation: The automatic translation of text or speech from one language to another by a computer. Chatbot: A computer program designed to simulate conversation with human users, often used in customer service or information retrieval.

	Sentiment Analysis: The process of determining the emotional tone behind a series of words, often used in text analysis to understand opinions or feelings. Voice Command: A verbal instruction given to a device to perform a specific action, such as making a call or playing a song. Artificial Intelligence (AI): The simulation of human intelligence processes by machines, especially computer systems, including learning, reasoning, and self-correction.
Activator:	<u>SMART START Activity (slide 2)</u> • Ask students: "When was the last time you controlled a computer with your voice? What did you ask it to do?" • Create a list of voice-controlled devices on the board (Siri, Gemini, Alexa, Copilot, ChatGPT). • Discuss students' experiences with these devices. • Explore different types of language interactions with computers (e.g., speech recognition, speech generation) (slides 4 - 6)
During the Learning:	(slide 7) Video - How Speech Recognition Works (slides 8 - 11) Scenarios Discussion: Present the following scenarios to the class and ask the students to identify the inputs, outputs, and whether they are examples of recognition or generation: • Scenario 1: Tonya asks her phone, "Who is the artist for the song that is playing?" • Scenario 2: A student talks into Google Translate to ask a question. • Scenario 3: A parent yells for Google Nest to search how to limit screen time. Questions to Discuss:

	1. What is the input in this scenario? Is it recognition or generation? 2. How was a specific technology used in this scenario? 3. What is the output in this scenario? Is it recognition or generation?
Summary/Reflection:	Reflection: Create a Journal Entry: Write a journal entry that captures your thoughts and experiences related to the following prompts: • Describe a time when you used a voice-controlled device (e.g., Siri, Alexa). What did you ask it to do, and how did it respond? • Reflect on how you think speech recognition and speech generation technologies have changed the way we communicate with computers. • Imagine a future application of AI that could improve daily life. What would it be, and how would it interact with language?
Assessment:	Have students write a short paragraph explaining how they use language to interact with a computer in their daily lives, including examples of both speech recognition and speech generation.
Differentiation:	For Struggling Learners: <u>Visual Aids and Graphic Organizers:</u> • Provide graphic organizers to help them categorize different forms of language interaction with computers. For example, a Venn diagram to compare speech recognition and text generation. <u>Simplified Vocabulary Lists:</u> • Create a list of key vocabulary with definitions and visuals. This can help them understand the terms used in discussions and activities. For Gifted Learners: <u>Independent Research Project:</u> • Allow gifted students to choose a specific area of AI language interaction (e.g., ethical implications of AI,

	advancements in NLP) for a deeper research project. They can present their findings to the class. <u>Create a More Complex Chatbot:</u> • Challenge them to design a chatbot with advanced features, such as incorporating sentiment analysis or multiple language support. They can present their chatbot and explain its functionalities.
Extension Activities:	<u>Explore AI Applications in Different Languages:</u> Encourage students to investigate how AI technologies handle multiple languages. They can compare the effectiveness of translation services like Google Translate in different languages and present their findings to the class. <u>Guest Speaker Session:</u> Arrange for a guest speaker who works in the field of artificial intelligence or natural language processing to discuss real-world applications of these technologies. Students can prepare questions in advance.